UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 006 · Problems on Euler's theorem · compiled 14 October 2026

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1 Short notes

1 · Testing homogeneity: the scaling test

Scaling test: $u(\lambda x, \lambda y) = \lambda^n u(x,y)$ means degree n (positive, negative or zero). Slide examples: $\frac{x^4+2x^2y^2+y^4}{(x^2+y^2)^2}$ degree 0; $\frac{\sin^{-1}(y/x)}{\sqrt{x^2+y^2}}$ degree -1; $\frac{x^2y+xy^2}{(x+y)^3}$ degree 0; $\frac{e^x-e^y}{x-y}$ not homogeneous.

2 · Verifying Euler's theorem directly

To verify: find n, differentiate, form $xu_x + yu_y$, simplify to nu. For $Z = \frac{x+y}{\sqrt{x}+\sqrt{y}}$ ($n = \frac{1}{2}$): $xZ_x + yZ_y = \frac{(x+y)(\sqrt{x}+\sqrt{y})}{2(\sqrt{x}+\sqrt{y})^2} = \frac{1}{2}Z$.

3 · When u itself is homogeneous: first and second order

u homogeneous of degree n: $xu_x + yu_y = nu$, $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$, sum n^2u. $\frac{x-y}{x^2+y^2}$: $-u$, $2u$. ESA: $5u$ & $20u$; $9u$; $17\pi/2$; ratios only give 0.

4 · When f(u) is homogeneous: the extension

If $f(u)$ has degree n: $g = nf/f'$, second order $g(g'-1)$. $e^{x^3y^3/(x^2+y^2)}$: $4u\log u$. $\log\frac{x^2+y^2}{x+y}$: 1, -1. $\sin^{-1}\frac{x+y}{\sqrt{x}+\sqrt{y}}$: $\frac{1}{2}\tan u$, $\frac{1}{2}\tan u(\frac{1}{2}\sec^2 u - 1)$. $\sin^{-1}\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$: $-3\tan u$. ESA: $\tan^{-1}\frac{x^4+y^4}{xy}$ gives $\sin 2u$, $\sin 2u(1-4\sin^2 u)$.

2 Keyword index

16 terms; the number in brackets is the topic.

Chain rule step (4). $F_x = f'(u)u_x$; then divide by $f'(u)$.

Degree (1). The power n of λ; positive, negative, fractional or zero.

Degree 0 (1). Depends on ratios only; Euler sum 0.

Euler's theorem (3). $xu_x + yu_y = nu$.

Extension (4). $xu_x + yu_y = nf(u)/f'(u)$ when $f(u)$ is homogeneous.

g(g′−1) (4). Second-order sum for the extension.

g(u) (4). $nf(u)/f'(u)$.

Not homogeneous (1). No single λ^n factors out (e.g. $e^x - e^y$).

n²u (3). First plus second order.

Quotient degree (1). Numerator degree minus denominator degree.

Quotient rule (2). Used to differentiate Z on slide 006·3.

Scaling test (1). $u(\lambda x, \lambda y) = \lambda^n u$.

Second-order corollary (3). $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$.

Sum of homogeneous parts (3). Each part with its own degree.

Undo the outer function (4). $\sin^{-1} \to \sin$, $\log \to e^u$, $e \to \log$, $\tan^{-1} \to \tan$.

Verification (2). Compute $xu_x + yu_y$ directly and show it equals nu.

3 Rapid recall

1	Scaling test for degree n?	$u(\lambda x, \lambda y) = \lambda^n u(x, y)$.
2	Degree of $\frac{x^4+2x^2y^2+y^4}{(x^2+y^2)^2}$?	0 (it is identically 1).
3	Degree of $\frac{\sin^{-1}(y/x)}{\sqrt{x^2+y^2}}$?	-1.
4	Degree of $\frac{x^2y+xy^2}{(x+y)^3}$?	0.
5	Is $\frac{e^x-e^y}{x-y}$ homogeneous?	No.
6	Degree of $\frac{x+y}{\sqrt{x}+\sqrt{y}}$?	$\frac{1}{2}$.
7	$u = \frac{x-y}{x^2+y^2}$: $xu_x + yu_y$?	$-u$.
8	Same u: second order?	$2u = \frac{2(x-y)}{x^2+y^2}$.
9	Extension formula?	$xu_x + yu_y = nf(u)/f'(u) = g(u)$.
10	Extension, second order?	$g(u)[g'(u) - 1]$.
11	$u = e^{x^3y^3/(x^2+y^2)}$: $f(u)$ and n?	$\log u$, $n = 4$.
12	Same u: $xu_x + yu_y$?	$4u \log u$.
13	$u = \log \frac{x^2+y^2}{x+y}$: both sums?	1 and -1.
14	$u = \sin^{-1} \frac{x+y}{\sqrt{x}+\sqrt{y}}$: $xu_x + yu_y$?	$\frac{1}{2} \tan u$.
15	Same u: second order?	$\frac{1}{2} \tan u(\frac{1}{2} \sec^2 u - 1)$.
16	$u = \sin^{-1} \frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$: $xu_x + yu_y + zu_z$?	$-3 \tan u$.
17	$u = \tan^{-1} \frac{x^4+y^4}{xy}$: both sums?	$\sin 2u$; $\sin 2u(1 - 4\sin^2 u)$.
18	$u = x^3y^2 \sin^{-1}(y/x)$: both sums?	$5u$; $20u$.
19	$x^3 \sin^{-1} \frac{y}{x} + x^4 \tan^{-1} \frac{y}{x}$, first + second at $(1, 1)$?	$17\pi/2$.
20	$f(x/y, y/z, z/x)$: Euler sum?	0.

Mnemonic (study aid). *"Undo, test, divide."* Undo the outer function to get $f(u)$, test $f(u)$ for its degree n, then divide: $g = n\,f/f'$.

4 Last 10 minutes before the exam

Top facts

1. Scaling test: $u(\lambda x, \lambda y) = \lambda^n u$; degree may be positive, negative or zero.
2. No single λ^n factors out $\Rightarrow$ not homogeneous ($\frac{e^x - e^y}{x-y}$).
3. u homogeneous: nu, $n(n-1)u$, together $n^2 u$; parts of different degrees separately.
4. $\frac{x-y}{x^2+y^2}$: $n = -1$, answers $-u$ and $+2u$.
5. $f(u)$ homogeneous: $g = nf/f'$; second order $g(g' - 1)$.
6. $\sin^{-1} \to n \tan u$; $\tan^{-1} \to \frac{n}{2} \sin 2u$; $\log \to n$ (second order $-n$); $e^{(\cdot)} \to nu \log u$.
7. Slide 006·4: $-3 \tan u$; slide 006·12: $4u \log u$; slide 006·14: $1, -1$; slide 006·16: $\frac{1}{2} \tan u$.
8. ESA favourites: $5u$ & $20u$; $\sin 2u$ & $\sin 2u(1 - 4\sin^2 u)$; $9u$; $17\pi/2$.

Top formulas

E1.1 · Scaling test

ESA ★★

$$u(\lambda x, \lambda y) = \lambda^n \, u(x, y) \implies u \text{ homogeneous of degree } n$$

GIVES	Whether u is homogeneous, and its degree.
APPLIES	Any function; with three variables scale z too.
NOTE	If no power λ^n factors out, u is not homogeneous.

S006·5 · T006b 00:00:52

E2.1 · Euler's theorem

ESA ★★★

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n \, u$$

GIVES	The first-order Euler sum.
APPLIES	u homogeneous of degree n; add $z \, u_z$ for three variables.
NOTE	To verify, compute the left side by differentiating and show it equals nu.

S006·3 · S006·10 · T006a 00:00:00

E3.1 · Second order

ESA ★★★

$$x^2 u_{xx} + 2xy \, u_{xy} + y^2 u_{yy} = n(n-1) \, u$$

GIVES	The second-order Euler sum.
APPLIES	u homogeneous of degree n.
NOTE	$n = -1$ gives $2u$; $n = -\frac{2}{3}$ gives $\frac{10}{9}u$.

S006·11 · T006b 00:07:23

E3.2 · First plus second order

ESA ★★★

$$x^2 u_{xx} + 2xy\, u_{xy} + y^2 u_{yy} + x\, u_x + y\, u_y = n^2 u$$

GIVES The combination asked in ESA Jan 2024 sets 3 and 4.

APPLIES u homogeneous of degree n; for $u_1 + u_2$ use $n_1^2 u_1 + n_2^2 u_2$.

NOTE $n(n-1) + n = n^2$.

ADDED FOR CLARITY ESA Jan-24 s3 1c · ESA Jan-24 s4 1d · CW U1 P23

E4.1 · Extension, first order

ESA ★★★

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\frac{f(u)}{f'(u)} = g(u)$$

GIVES The Euler sum when u is not homogeneous but $f(u)$ is.

APPLIES $f(u)$ homogeneous of degree n; add $z\, u_z$ for three variables.

NOTE $\sin \to n \tan u$, $\tan \to \frac{n}{2} \sin 2u$, $e^u \to n$, $\log \to nu \log u$.

S006·13 · S006·15 · S006·17 · T006b 00:09:08

E4.2 · Extension, second order

ESA ★★★

$$x^2 u_{xx} + 2xy\, u_{xy} + y^2 u_{yy} = g(u)\left[g'(u) - 1\right]$$

GIVES The second-order sum for the extension.

APPLIES g from E4.1; differentiate g with respect to u.

NOTE If g is a constant (the $\log$ case), the answer is $-g$.

S006·15 · S006·17 · T006b 00:10:59